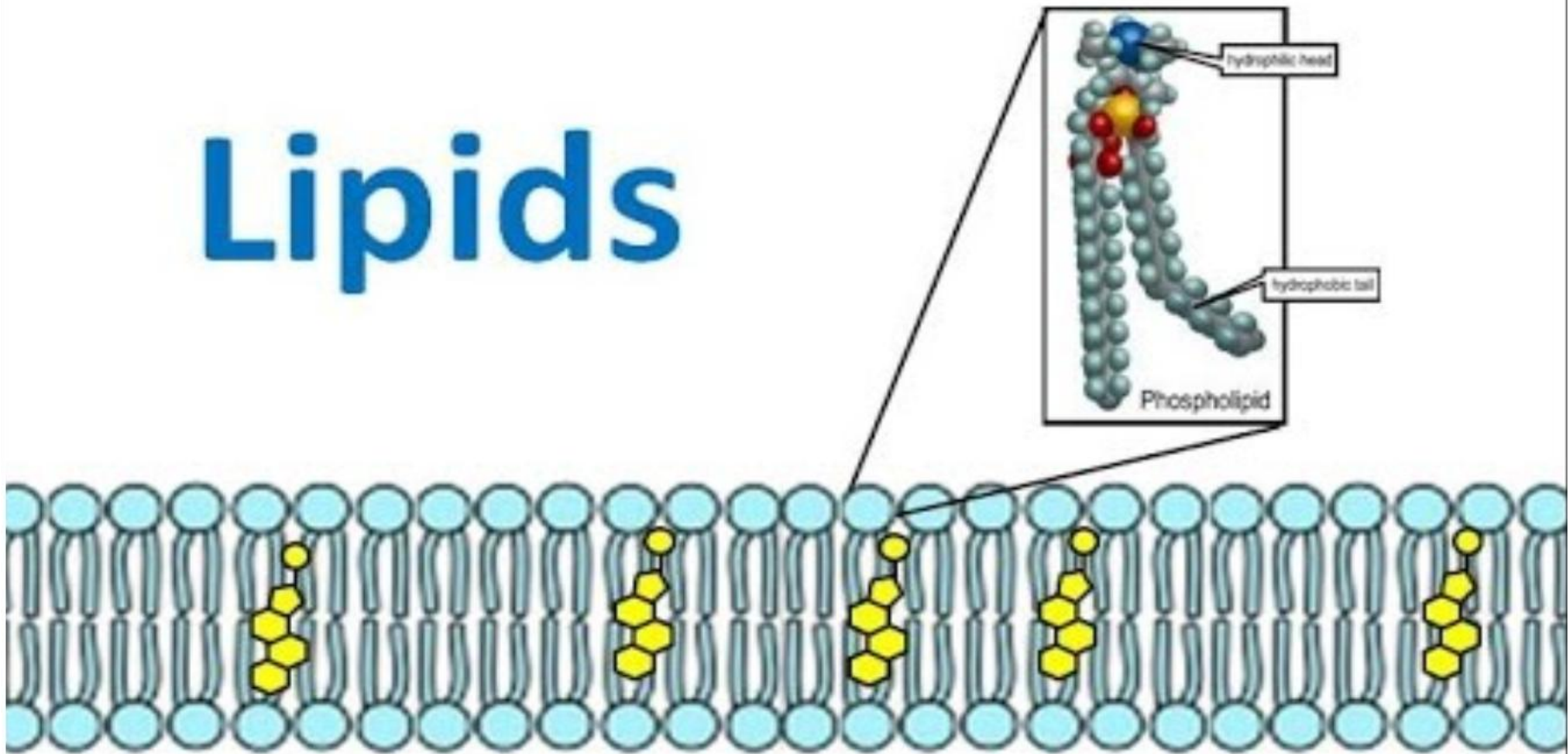


Lipids

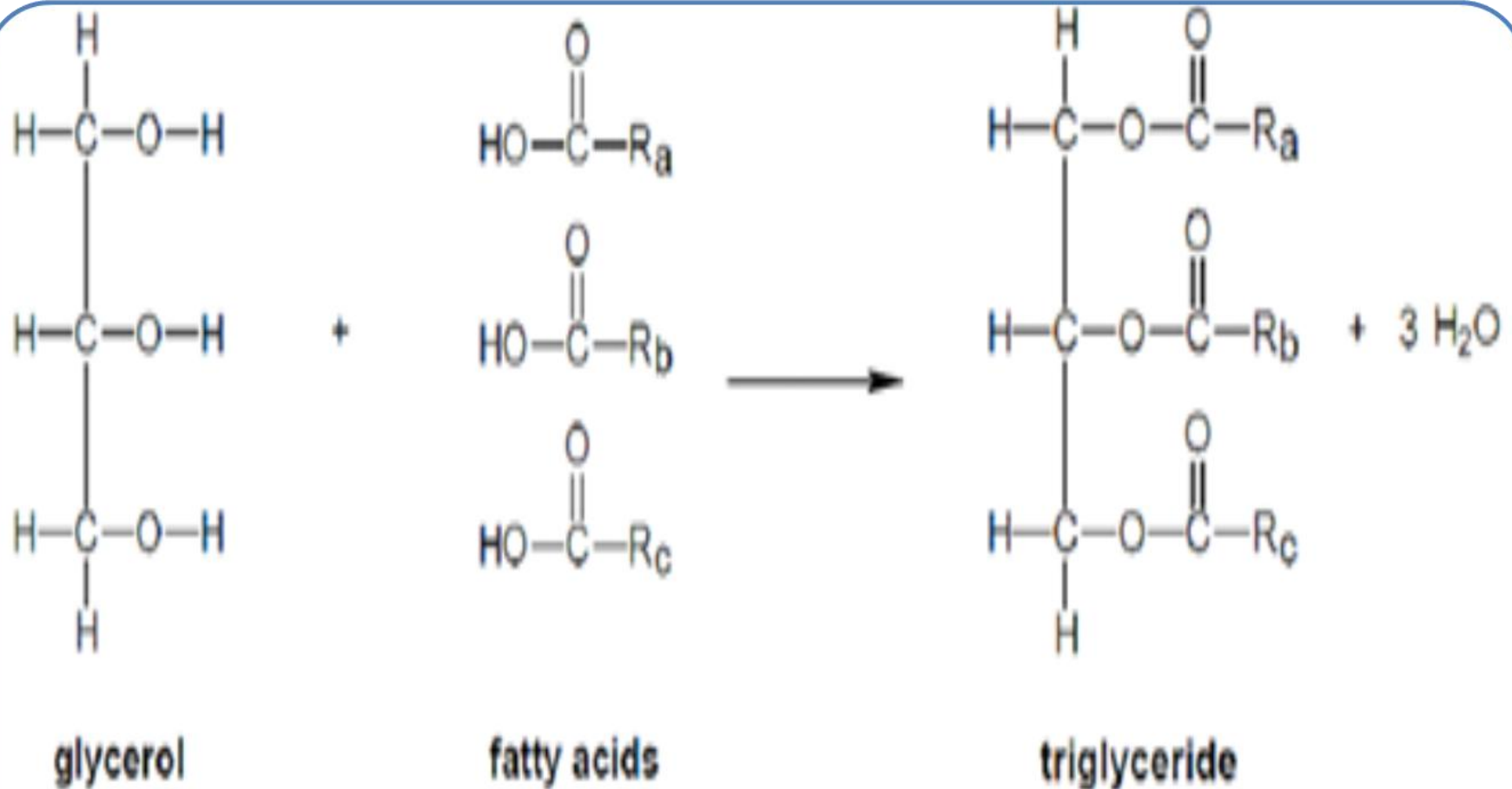


LIPIDS

These are large group of organic compounds. Like carbohydrates, they contain carbon, hydrogen and oxygen but the proportion of oxygen is smaller than in carbohydrates hence they are more reduced than the carbohydrates.

- ❖ Lipids are **insoluble** in water.
- ❖ They are of two types i.e. **fats and oils**. Fats are solid at room temperature while oils are liquids at room temperature.
- ❖ Lipids are made of fatty acids and glycerol. **Glycerol** has **3 OH groups** and each combines with a separate fatty acid to form a lipid chemically known as a **triglyceride**.
- ❖ This is a **condensation reaction** that leads to liberation of 3 water molecules.

Formation of a triglyceride



Glycerol + 3 fatty acids → triglyceride

FATTY ACIDS

- ❖ All occurring lipids have **glycerol** and therefore it is the nature of the fatty acids which determines the characteristics of any particular lipid.
- ❖ All fatty acids have **a carboxyl group** (COOH), the remainder of the molecule being a hydro carbon chain of varying length.
- ❖ These chains may possess one or more **double bond** in which case it is said to be **unsaturated**. If it possesses no double bonds, it is said to be **saturated**.

Fatty acids

Nature of fatty acid	General formula	Saturated/unsaturated	Occurrence
• Butyric acid	C_3H_7COOH	Saturated	<i>Butter fat</i>
• Linoleic acid	$C_{17}H_{31}COOH$	Unsaturated	<i>Seed oil</i>
• Oleic acid	$C_{17}H_{33}COOH$	Unsaturated	<i>All fats</i>
• Palmitic acid	$C_{15}H_{31}COOH$	Saturated	<i>Animal & veg fat</i>
• Stearic acid	$C_{25}H_{51}COOH$	Saturated	<i>Wood oil</i>
• Arachidic acid	$C_{19}H_{39}COOH$	Saturated	<i>P.nut oil</i>

From the table, it is seen that the ***hydrocarbon chains*** may be very long forming long tails which extend from the glycerol molecules.

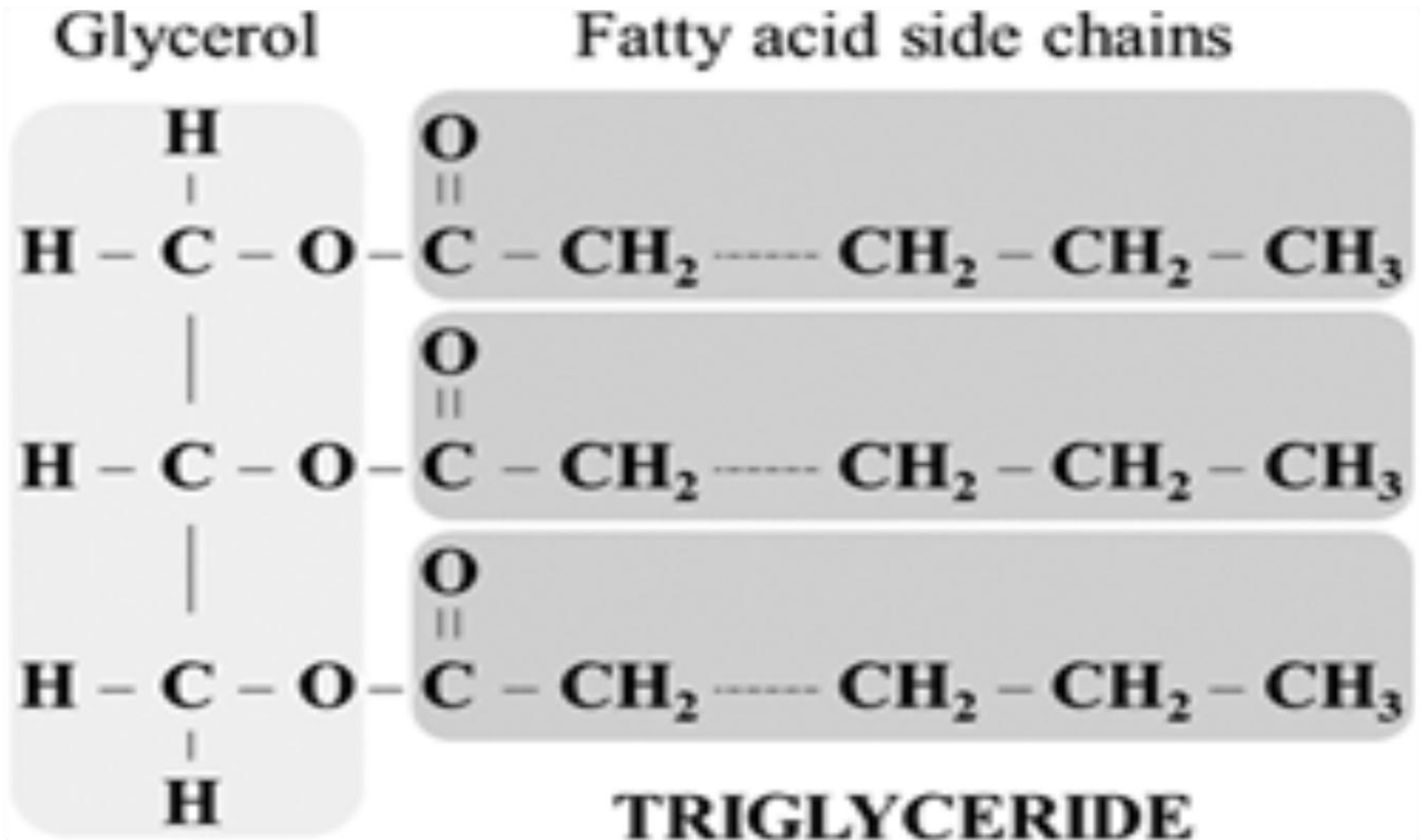
These tails are hydrophobic (water repelling) which makes the lipids insoluble in water.

Question: *explain why lipids are insoluble in water?*

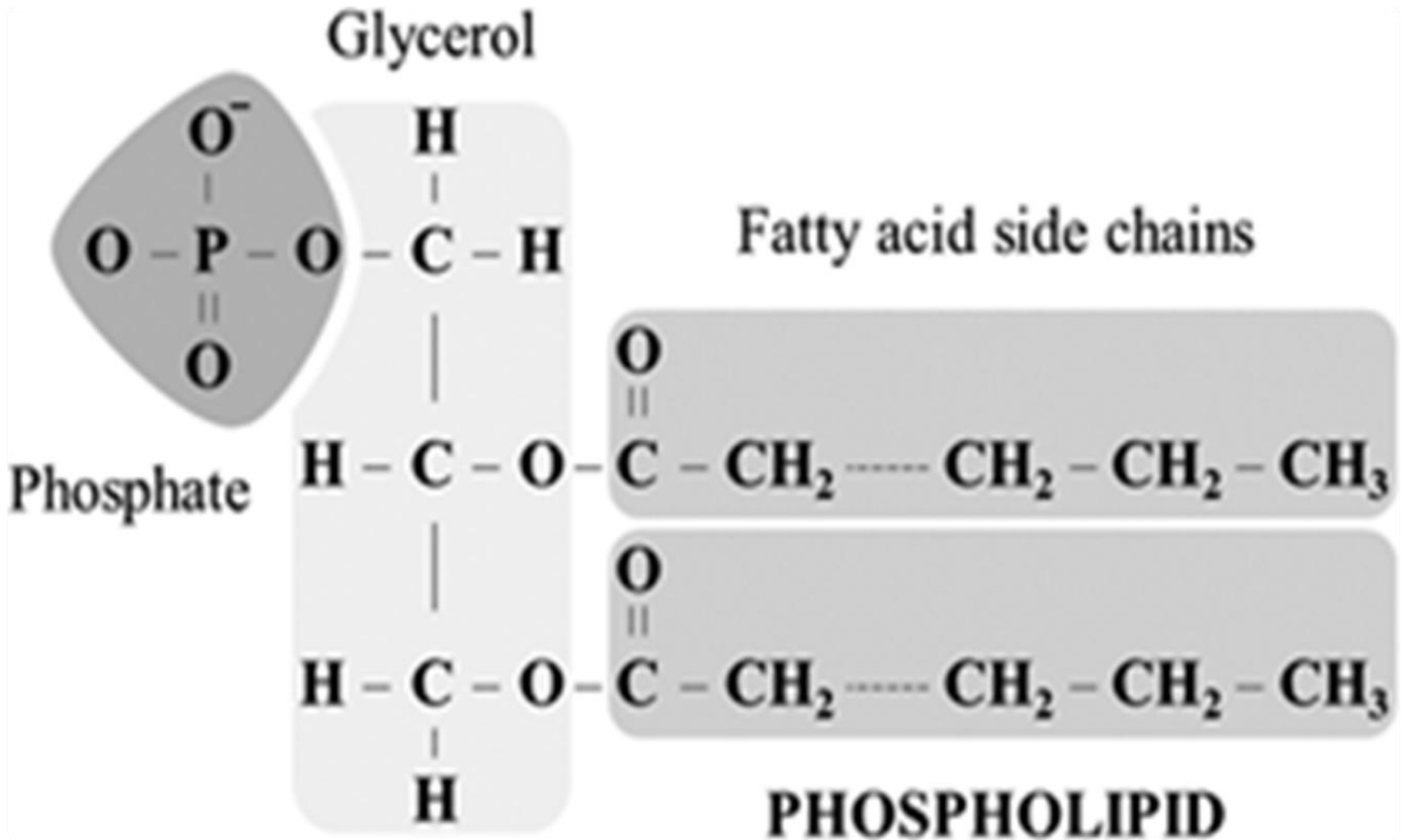
PHOSPHOLIPIDS

These are lipids in which one of the fatty acid groups is replaced by phosphoric acid.

Remember formation of triglyceride



Phospholipid structure



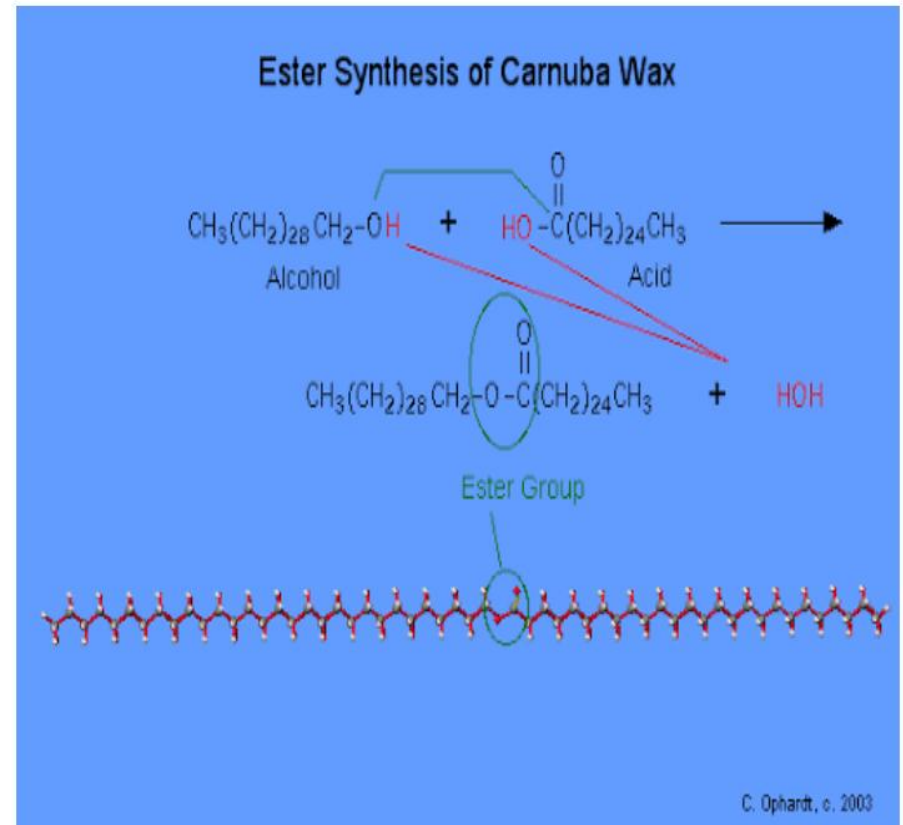
- **Phosphoric acid** is **hydrophilic** in contrast to the remainder of the molecule, having an end attracting water while the **other end repelling** it gives the phospholipid its characteristics as one the components of the cell membrane.

Credit : Sr. Florence Kajoina

WAXES

These are formed by combination **of fatty acids** with an ***alcohol other than glycerol***.

Their major function is water proofing in plants and animals. They are also storage compounds in seeds e.g. castor oil seeds.

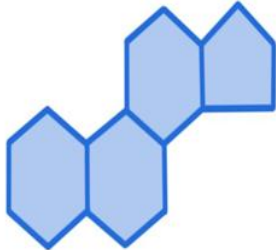
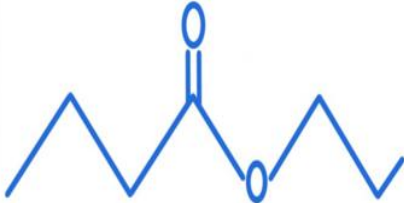


STEROIDS

These are related to lipids e.g. cholesterol in animals used to synthesize **sex hormones**.

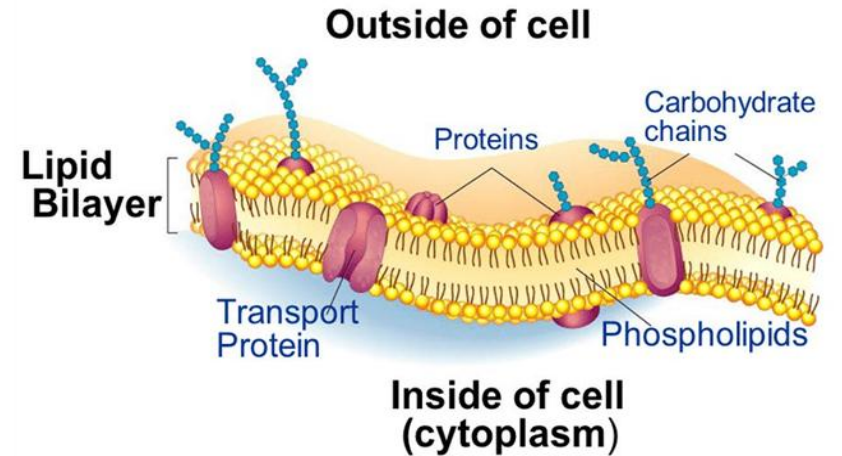
Other steroids include **vitamin B** and **bile acids**.

Types of Lipids: Steroids and Waxes

STEROIDS	WAXES
Steroids are characterized by a 4 ring structure.  Steroids are found in the cell membrane, where they influence the membrane's fluidity. They can also act as hormones in the body.	Waxes are esters made of an alcohol chain and a fatty acid chain.  Many plants have wax-covered leaves to prevent water loss.

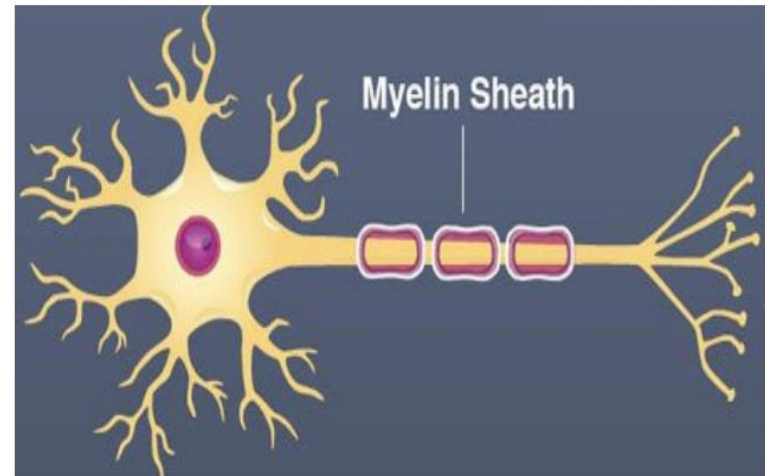
STRUCTURAL FUNCTIONS OF LIPIDS

1. They are components of the ***plasma/cell membrane***.
2. They form **subcutaneous fat** in the dermis of the skin hence insulating the body since they are poor conductors of heat.



STRUCTURAL FUNCTIONS OF LIPIDS

3. They are components of the **waxy cuticle** in plants and insects there by preventing water loss (desiccation).
4. They form a component of the **myelin sheath** of nerves hence playing a role in the transmission of impulses.



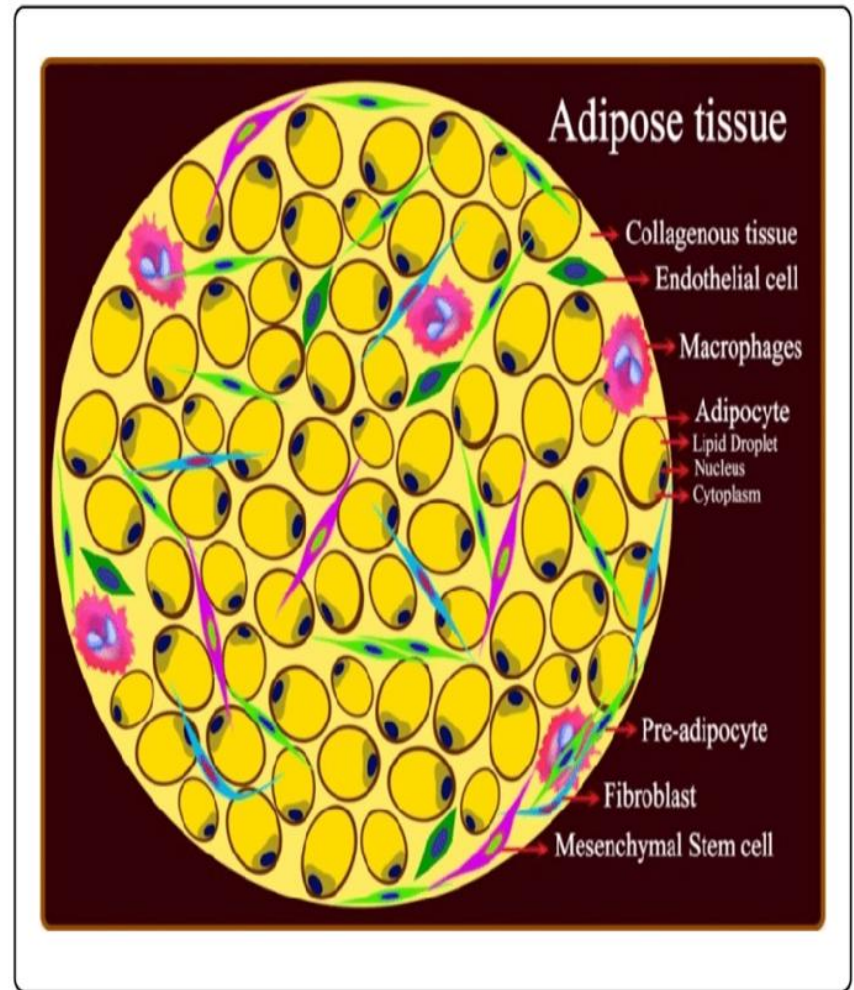
STRUCTURAL FUNCTIONS OF LIPIDS

- 5. They **protect delicate organs** e.g. the heart and kidney from injury.
- 6. They **coat on fur** of animals enabling it to **repel water** which would otherwise wet the organism.



STRUCTURAL FUNCTIONS OF LIPIDS

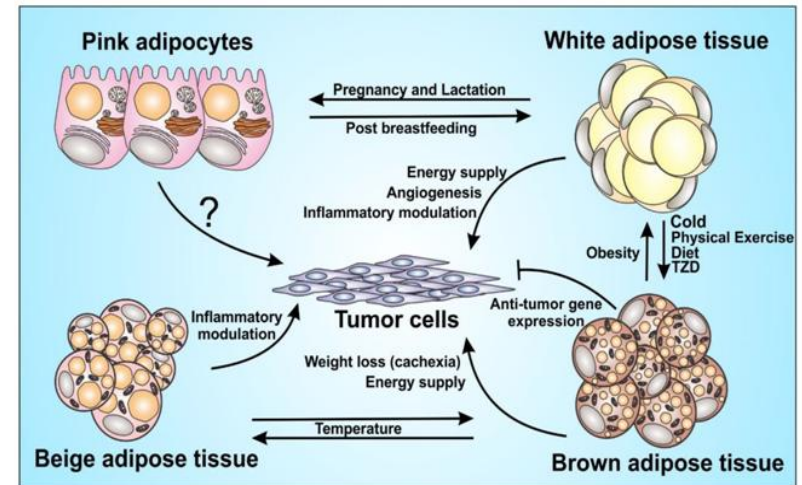
7. They are component of **adipose** tissue.



Physiological functions of Lipids

iii) They are a good source of **metabolic water** to desert animals, young birds and reptiles while still in their shells.

iv) They are a constituent of the **brown adipose tissue** which provides heat for temperature regulation (thermogenesis).



Other functions of lipids

- i) Some lipids provide a **scent** in plants which attracts insects for **pollination**.
- ii) Wax is used by bees to construct honey combs.



k59320966 www.fotosearch.com

Other functions of lipids

iii) Wax
from bees
is used in
the
manufactu
re of
candles



QUESTION:

***WHAT PROPERTIES DO LIPIDS
POSSES AS STORAGE
COMPOUNDS?***

Solution

- ❖ They are **compact** taking up little space.
- ❖ They are **insoluble** in water hence cannot be lost in solution.
- ❖ They are **light** to keep the weight to a minimum and allow buoyancy.
- ❖ They have a high **calorific energy value**.
- ❖ They have a high **hydrogen-oxygen** content hence can yield a lot of water on oxidation.

TESTS FOR LIPIDS

They are tested for using the;

- ❖ *Sudan III dye test,*
- ❖ Ethanol **emulsion test** or
- ❖ The grease spot
(translucent spot) test.

Sudan III test:

Procedure	Observation	Deduction
To 1cc of food solution, add 1cc of Sudan III and shake.	A turbid solution turns a red emulsion.	Lipids present.
	Turbid or colourless solution remains a turbid or colourless solution.	Lipids absent.

The emulsion test:

The reagents used are **ethanol** and **water**.

Procedure	Observation	Deduction
<i>To 1 cc of food solution, add 1 cc of ethanol and shake. Then add 5 drops of water and shake.</i>	A turbid solution turns to a cream emulsion	Lipids present.
	Turbid or colourless solution remains a turbid or colourless solution.	Lipids absent.

Ethanol emulsion test

[https://www.youtube.com/watch?v= BcXylW7JFc](https://www.youtube.com/watch?v=BcXylW7JFc)

Translucent spot test:

Procedure	Observation	Conclusion
<i>Add 2 drops of test solution on a piece of filter paper.</i> <i>Allow to dry and observe under light.</i>	A translucent spot or patch is left on the paper.	Lipids present
	No translucent spot is formed on the paper.	Lipids absent.

A permanent translucent paper test

